

Verifier-Guided Structured Visual Test-Time Scaling for Multimodal Reasoning

Anonymous ACL submission

Abstract

Multimodal reasoning questions often admit fluent textual rationales that are not visually supported, making longer chain-of-thought an unreliable default for vision-language models. Recent work on chain-of-thought, self-consistency, visual scratchpads, and multimodal verification motivates a narrower question: under a matched inference budget, can structured visual evidence candidates help a model select answers on real multimodal benchmarks? We study candidate-generation repair: the controller starts from a direct-answer seed, adds family-specific visual candidates, and applies a predeclared majority-first no-gold selector before official scoring. The paper-facing evaluation targets MathVista, MMMU, MATH-Vision, and MMStar with direct-answer, vanilla-CoT, matched-budget self-consistency, answer-seed floor, fixed-family selector, priority-selector ablation, and majority-first proposed conditions. Final comparative accuracy, matched-budget deltas, and scope boundaries are intentionally reserved as [TBD_ABSTRACT_HEADLINE_ACCURACY], [TBD_ABSTRACT_MATCHED_BUDGET_DELTA], and [TBD_ABSTRACT_SCOPE_BOUNDARY]; no result claim in this draft should be read as complete until those markers are backfilled from terminal run artifacts.

1 Introduction

Textual chain-of-thought is a useful reasoning interface for language models, but visual reasoning imposes a different failure mode: a model can narrate a plausible path while ignoring the image evidence that should constrain the answer (Wei et al., 2022; Zhang et al., 2023; Kancheti et al., 2026). Self-consistency improves many text-reasoning tasks by spending more samples and voting (Wang et al., 2023), but vote aggregation alone does not check whether the winning answer is grounded in visual structure. The motivating problem for this paper is

therefore selective visual test-time scaling: spend extra inference only when the extra candidate evidence is likely to make the answer more visually faithful.

The operator-specified setting is deliberately constrained. The evaluated benchmarks are real public multimodal benchmark families, not synthetic local episodes: MathVista (Lu et al., 2024), MMMU (Yue et al., 2024a), MATH-Vision (Wang et al., 2024), and MMStar (Chen et al., 2024). The core baselines are direct answer, vanilla text chain-of-thought, and matched-budget self-consistency. The proposed system must compete at comparable inference budget and must not rely on gold answers, exact-oracle selection, or post-hoc row filtering.

Our working thesis is that structured visual chain-of-thought should be treated as an evidence representation rather than as the result itself. The system asks an open vision-language model to produce multiple answer candidates, including a direct-answer seed and family-specific visual candidates that explicitly transcribe labels, table cells, coordinate state, option-level constraints, and terminal answer surfaces. A no-gold selector then chooses among candidates using answer agreement, visual-support cues, and answer-surface checks that are available before scoring. This separates the candidate-generation question from the selection question, and it gives the paper a concrete way to test whether visual structure helps beyond simply asking for longer rationales (Hu et al., 2024; Menon et al., 2024; Sun et al., 2025).

The draft currently makes no positive empirical claim. The final Introduction should preview the main result only after [TBD_INTRO_HEADLINE_FINDING] and the main failure boundary only after [TBD_INTRO_FAILURE_RATE] are populated from the terminal full-matrix audit and the later results table. Until then, the contribution is a scaffolded, evidence-disciplined paper plan: a structured

controller, a matched-budget benchmark protocol, and explicit markers for every result-dependent statement.

2 Related Work

Reasoning traces and test-time scaling. Chain-of-thought prompting established intermediate text rationales as a practical interface for eliciting reasoning from large language models (Wei et al., 2022). Self-consistency then showed that sampling multiple rationales and aggregating answers can improve reasoning outcomes in text settings (Wang et al., 2023). More recent multimodal test-time scaling work extends this idea to multimodal models and iterative refinement (Chen et al., 2026). Our paper keeps the matched-budget scaling premise but asks whether the selected candidate is visually grounded, not only frequent.

Visual reasoning traces. Multimodal chain-of-thought and visual scratchpad methods argue that some tasks benefit from externalizing intermediate visual state (Zhang et al., 2023; Hu et al., 2024; Menon et al., 2024; Shi et al., 2025). These papers motivate structured visual evidence, but they also make a simple always-use-visual-CoT story insufficient: if the visual trace is wrong or unnecessary, extra structure can amplify the wrong evidence. The current draft therefore treats visual candidates as alternatives that must be verified rather than as a guaranteed improvement.

Verification and process supervision. Process-supervision work in mathematical reasoning shows that supervising intermediate steps can be more informative than outcome-only supervision (Lightman et al., 2023). Multimodal verification and process-reward papers adapt that intuition to vision-language reasoning (Sun et al., 2025; Wu et al., 2026; Zhang et al., 2025). Our method follows the verification direction, but its paper-facing claim is restricted to the exact controller and benchmark protocol once the run evidence is complete.

Benchmark pressure. MathVista, MMMU, MATH-Vision, and MMStar stress complementary forms of visual mathematical, expert, and perception-reasoning ability (Lu et al., 2024; Yue et al., 2024a; Wang et al., 2024; Chen et al., 2024). MMMU-Pro and related benchmark audits further emphasize that multimodal evaluation should guard against text-only shortcuts and brittle answer extraction (Yue et al., 2024b). This paper uses the

benchmark mix to test whether structured evidence transfers across families rather than only repairing one MathVision-style answer surface.

3 Method

Let a task instance contain an image, a question, optional answer choices, and benchmark-specific scoring rules. The controller produces a small candidate set under a fixed inference budget. The paper-facing method name will be locked as `[TBD_METHOD_NAME_LOCK]`; the current run evaluates candidate-generation repair with a direct-answer seed, family-specific visual candidates, and the proposed selector policy recorded internally as `majority_first_no_gold`.

Candidate families. The direct-answer family emits a concise final answer. The vanilla-CoT family emits a text rationale followed by a final answer. The structured visual-evidence family asks the model to construct candidates that isolate visible facts before deciding: direct-answer seeding, diagram label and scale transcription, symbolic table or pattern reconstruction, coordinate-state tracking, option-level visual elimination, family-specific benchmark repair, and terminal answer repair. These families are motivated by the failure audits in the local research artifacts, but the draft defers any claim about which family wins to `[TBD_ANALYSIS_BY_BENCHMARK]`.

Verifier-guided selection. The selector scores each candidate by paper-facing criteria: whether the final answer can be parsed by the official scorer, whether cited visual evidence is specific to the input, whether numeric or spatial statements are unsupported, whether the candidate respects answer choices, and whether the answer surface is compatible with the benchmark. The active proposed selector is `majority_first_no_gold`; it is predeclared before the current-scale run and cannot inspect gold answers, row scores, task identifiers, source task identifiers, split labels, post-hoc correctness, or oracle outputs. This draft reserves the final verifier/training/adaptation statement as `[TBD_VERIFIER_TRAINING_OR_ADAPTATION]` until the paper names the component consistently with the terminal artifacts.

Budget matching. The main comparison is not “more compute beats less compute.” The proposed condition is compared against direct answer, vanilla-CoT, matched-budget self-

consistency, answer-seed floor, fixed-family selector, and the priority-selector ablation with matched generation count, maximum completion budget, and scorer access where applicable. The exact model/backend and decoding budget are fixed in the experiment manifest and will be stated as `[TBD_EVALUATED_MODEL_BACKEND]` and `[TBD_DECODING_BUDGET]`.

4 Experimental Setup

Benchmarks. The evaluation targets MathVista, MMMU, MATH-Vision, and MMStar. Each benchmark is represented by official task identifiers and local provenance records. The active scale-up contract expects 1,680 terminal rows: four benchmark families, 60 official-source tasks per family, and seven conditions. The exact splits, filtering rules, terminal row checks, and any benchmark-specific exclusions will be filled as `[TBD_DATASET_COUNTS]` after the terminal audit confirms the final run matrix.

Models and framework. Open-weight VLM inference is routed through the project evaluation wrapper and the selected framework-backed inference stack. The active run uses Qwen2.5-VL-family inference as the open VLM backbone (Bai et al., 2025); final paper wording will name the precise evaluated model, framework, and decoding settings only after `[TBD_EVALUATED_MODEL_BACKEND]` is checked against the manifest.

Baselines and metrics. The required conditions are direct answer, vanilla-CoT, matched-budget self-consistency, answer-seed floor, fixed family-prior selector, priority-selector ablation, and the proposed majority-first candidate-generation repair selector. The proposed condition uses candidate-generation repair with a direct-answer seed plus family-specific visual candidates. Primary metric is official exact-match or benchmark-provided accuracy after official answer extraction; any scorer-specific normalization will be summarized as `[TBD_SCORING_PROTOCOL]`. Completion status for the full matrix is `[TBD_FULL_MATRIX_COMPLETION]`.

5 Results

This section is intentionally a result scaffold. Table 1 names the final comparison cells, but every

numeric entry is a draft marker until the terminal run artifacts and result table are generated.

The completed Results prose will answer three questions. First, does the proposed controller improve the matched-budget aggregate over the strongest baseline? Second, are gains concentrated in a particular benchmark family or do they transfer? Third, does the latency or sampling cost alter the interpretation of any accuracy change? The final claims for these questions are reserved as `[TBD_OVERALL_DELTA]`, `[TBD_PAIRED_SIGNIFICANCE]`, and `[TBD_COST_LATENCY]`.

6 Analysis and Ablations

The analysis will separate mechanism from aggregate score. The planned ablations remove visual structure, remove verifier-guided selection, and reduce candidate budget. Their final outcomes are reserved as `[TBD_ABLATION_NO_VISUAL_STRUCTURE]`, `[TBD_ABLATION_NO_VERIFIER]`, and `[TBD_ABLATION_HALF_BUDGET]`.

The benchmark-family analysis will check whether the method is only repairing MathVision answer surfaces or whether it also helps on MathVista, MMMU, and MMStar. That interpretation is reserved as `[TBD_ANALYSIS_BY_BENCHMARK]`. No ablation in this draft is claimed as positive, negative, or statistically reliable until the corresponding raw rows and result tables exist.

7 Failure Cases

The failure analysis will group errors by visually unsupported numbers, misread diagram labels, option-letter confusion, coordinate or transformation state errors, table or pattern reconstruction errors, and terminal answer extraction failures. The final taxonomy counts are reserved as `[TBD_FAILURE_TAXONOMY_COUNTS]`.

Qualitative examples will be included only if they are drawn from saved run rows with image/task provenance. The current draft reserves the example identifier, model output, and verifier decision as `[TBD_QUAL_EXAMPLE_ID]`, `[TBD_QUAL_EXAMPLE_MODEL_OUTPUT]`, and `[TBD_QUAL_EXAMPLE_VERIFIER_DECISION]`.

8 Conclusion

This draft frames structured visual reasoning as a candidate-evidence problem rather than as an un-

Benchmark	Metric	Direct	Vanilla-CoT	Self-consistency	Answer-seed floor	Fixed-family selector	Priority ablation	Majority-first proposed
MathVista	Accuracy	[TBD_MATHVISTA_DIRECT_ACC]	[TBD_MATHVISTA_COT_ACC]	[TBD_MATHVISTA_SC_ACC]	[TBD_MATHVISTA_ANSWER_SEED_FLOOR_ACC]	[TBD_MATHVISTA_FIXED_SELECTOR_ACC]	[TBD_MATHVISTA_PRIORITY_SELECTOR_ABLATION_ACC]	[TBD_MATHVISTA_OURS_ACC]
MMMU	Accuracy	[TBD_MMMU_DIRECT_ACC]	[TBD_MMMU_COT_ACC]	[TBD_MMMU_SC_ACC]	[TBD_MMMU_ANSWER_SEED_FLOOR_ACC]	[TBD_MMMU_FIXED_SELECTOR_ACC]	[TBD_MMMU_PRIORITY_SELECTOR_ABLATION_ACC]	[TBD_MMMU_OURS_ACC]
MATH-Vision	Accuracy	[TBD_MATHVISION_DIRECT_ACC]	[TBD_MATHVISION_COT_ACC]	[TBD_MATHVISION_SC_ACC]	[TBD_MATHVISION_ANSWER_SEED_FLOOR_ACC]	[TBD_MATHVISION_FIXED_SELECTOR_ACC]	[TBD_MATHVISION_PRIORITY_SELECTOR_ABLATION_ACC]	[TBD_MATHVISION_OURS_ACC]
MMSiir	Accuracy	[TBD_MMSTAR_DIRECT_ACC]	[TBD_MMSTAR_COT_ACC]	[TBD_MMSTAR_SC_ACC]	[TBD_MMSTAR_ANSWER_SEED_FLOOR_ACC]	[TBD_MMSTAR_FIXED_SELECTOR_ACC]	[TBD_MMSTAR_PRIORITY_SELECTOR_ABLATION_ACC]	[TBD_MMSTAR_OURS_ACC]

Table 1: Main matched-budget result matrix. The final caption must state the evidence-backed takeaway after [TBD_OVERALL_DELTA], [TBD_PAISED_SIGNIFICANCE], and [TBD_COST_LATENCY] are computed from canonical artifacts.

conditional chain-of-thought upgrade. The final conclusion will state the scoped empirical result only after [TBD_CONCLUSION_RESULT] is backed by the terminal run audit and results report. If the method does not beat the strongest matched-budget baseline, the conclusion will report that measured negative directly and focus on the mechanism diagnosis rather than relabeling the result.

Limitations

The current manuscript is not submission-ready because the full matrix and downstream analysis are still incomplete. The final paper must avoid broad claims about all multimodal reasoning, all VLMs, or all visual chain-of-thought methods. It will be limited to the evaluated benchmark families, the selected open VLM and configured API models, the exact inference budget, and the available ablations.

Ethical Considerations

The work studies benchmarked reasoning behavior and does not introduce a user-facing deployed system. Potential risks include overstating benchmark improvements, hiding failed comparisons, or producing rationales that appear more visually grounded than they are. The paper mitigates these risks by preserving direct baselines, matched-budget controls, failure cases, and explicit limitations. Dataset and model license/access notes must be verified before release packaging.

Reproducibility Checklist

The final checklist will report benchmark source versions, task counts, model/backend settings, decoding budgets, seeds or deterministic settings, run identifiers, artifact hashes, and scorer versions. The current draft reserves artifact hashes, final row counts, seed policy, and hardware/software summary as [TBD_RELEASE_ARTIFACT_HASHES], [TBD_CHECKLIST_FINAL_ROW_COUNTS], [TBD_CHECKLIST_SEEDS], and [TBD_CHECKLIST_HARDWARE].

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